SS–T–306b
14 JULY 1952
SUPERSEDING
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28 May 1943

FEDERAL SPECIFICATION

TILE, FLOOR, ASPHALT

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. CLASSIFICATION

1.1 Grade and type.—Asphalt floor tile covered by this specification shall be of one grade and type.

1.2 Thickness.—Asphalt tile shall be of the following thicknesses as specified in the invitation for bids:

⅛ inch
3/16 inch
¼ inch

1.3 Color and size.—The color and size shall be as specified or as agreed upon. (See 3.3, 6.1, 6.3 and 6.4.)

2. APPLICABLE SPECIFICATIONS

2.1 There are no other specifications applicable to this specification, except those referenced in section 7 by individual departments.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under "General Information" in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

3. REQUIREMENTS

3.1 Bid sample.—A sample of each color shall be submitted with the bid to show color, finish, and surface characteristics only.

3.2 Material.—The tile shall be composed of a thoroughly blended composition of thermoplastic binder of the asphaltic or resinous type, asbestos and other fibers, and inert materials as fillers or pigments, formed under pressure while hot and cut to size.

3.3 Color, finish, and surface characteristics.—The tile shall have a smoothly calendered wearing surface. The color, finish, and wearing surface shall match a sample mutually agreed upon by the buyer and seller. Tiles of plain colors shall be uniform in color throughout. In mottled and marbleized tiles, the marbleizing and mottling shall be worked throughout the full thickness of the material.

3.4 Size.—The tile shall be of the linear dimensions specified in the invitation for bids. (See 4.3.1, 6.1.1, and 6.3.)

3.4.1 *Tolerance.*—A tolerance of ±0.016 (1/64) inch in 12 inches shall be permitted.

3.5 Thickness.—The thickness of the tile shall be ⅛ inch, 3/16 inch, or ¼ inch, as specified in the invitation for bids, with a tolerance of ±0.005 inch. (See 4.3.2.)

3.6 Indentation.

3.6.1 *Indentation at 77° F.*

3.6.1.1 *One minute.*—When tested as described in 4.3.3.1, at the end of 1 minute the tile shall show an indentation of not less than 0.007 inch and not more than 0.018 inch.

3.6.1.2 *Ten minutes.*—When tested as described in 4.3.3.1, the indentation at the end of 10 minutes for ⅛-inch and 3/16-inch tile shall be as required in table I (or interpolated therefrom) and the indentation for ¼-inch tile shall be as given in table II (or interpolated therefrom), corresponding to the indentation recorded at the end of 1 minute.

3.6.2 *Indentation at 115° F.*—When tested as described in 4.3.3.2, ⅛-inch and 3/16-inch tile shall show an indentation of not more than 0.036 inch at the end of 30 seconds and ¼ inch tile shall show an indentation of not more than 0.042 inch at the end of 30 seconds.

3.7 Impact.—The tile shall not break or crack when subjected to the impact test specified in 4.3.4.

3.8 Deflection.—When tested as described in 4.3.5, ⅛-inch and 3/16-inch tile shall deflect

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and the average shall be taken as the thickness of the tile.

4.3.3 *Indentation.*

4.3.3.1 *Indentation at 77° F.*

4.3.3.1.1 *Specimen.*—The specimen shall consist of a single tile.

4.3.3.1.2 *Apparatus.*—The apparatus shall be as follows:

4.3.3.1.2.1 *Indentation tester.*—A McBurney[1] Indentation Tester (see fig. 1) shall be used for determining the indentation of the tile. The McBurney tester consists essentially of a rod with a hemispherical end 0.25 inch in diameter through which an initial load of 2 pounds and a final total of 30 pounds can be applied to the surface of the tile and a dial gage from which the indentation can be read.

4.3.3.1.2.2 *Constant-temperature bath.*—A constant-temperature water bath (see fig. 2) capable of maintaining the specimen at the required temperature within ±2° F.

4.3.3.1.2.3 A stop watch or other equivalent means for determining the time of application of the load.

4.3.3.1.2.4 A flat glass plate at least ¼ inch in thickness mounted in the water bath in such a manner that it will support the specimen and indentation tester so that the plate, specimen, and base of the indentation tester are fully immersed in the water.

4.3.3.1.3 *Procedure.*

4.3.3.1.3.1 The specimen, glass plate, and base of the indentation tester shall be conditioned for not less than 15 minutes by immersing in the water bath at a temperature of 77° ±1° F. The total time of conditioning and testing the specimen shall not exceed 90 minutes. The specimen, plate, and base of the indentation tester shall be conditioned for the same period of time and the indentation shall be determined with the specimen, plate, and base of the indentation tester fully immersed in water.

4.3.3.1.3.2 The specimen shall be placed on the glass plate and the base of the indentation tester placed on the surface of the tile. The hemispherical end of the rod of the tester shall be applied to the upper or wearing surface of the specimen under a load of 2 pounds. The tester shall be set so that a one-quarter turn of the load release handle will apply an additional load of 28 pounds to the rod. The dial shall be adjusted to zero and the additional load of 28 pounds (total 30 pounds) applied to the specimen within 5 seconds after adjusting the dial to zero. The readings of the dial at the end of 1 minute and at the end of 10 minutes from the application of the 28-pound load shall be recorded.

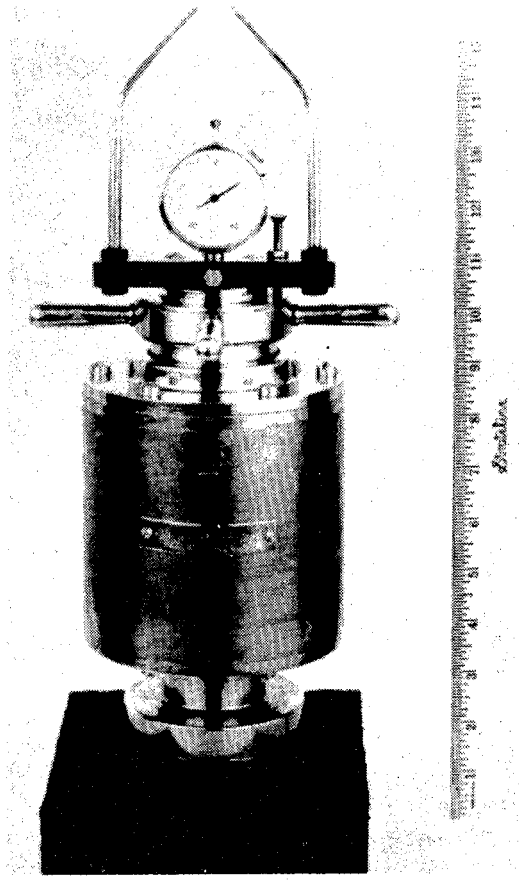


FIGURE 1.—*McBurney indentation tester.*

4.3.3.1.4 *Result.*

4.3.3.1.4.1 One-minute and 10-minute readings shall be taken at five scattered points on each specimen. The average of the five readings taken at each time period shall be the 1-minute and 10-minute indentation of the

[1] The "McBurney" tester is described in an article entitled "Indentation of Asphalt Tile" by J. W. McBurney, proceedings of the A. S. T. M. volume 34, part II, 1934, pages 591 to 603. Detail drawings of the instrument may be obtained free of charge from the National Bureau of Standards.

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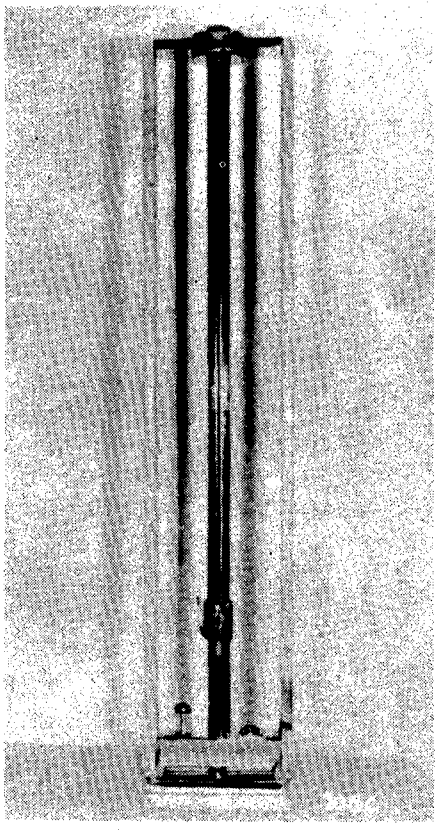


FIGURE 3.—*Impact-tester.*

4.3.4.2.3 *Constant-temperature bath.*—A constant-temperature bath as described in 4.3.3.1.2.2.

4.3.4.3 *Procedure.*

4.3.4.3.1 Two specimens shall be conditioned for not less than 15 minutes nor more than 30 minutes by immersion in the water bath at a temperature of 77° ±1° F. One specimen shall be removed from the water bath and a thin paste of zinc oxide in water immediately spread over the center of the wearing surface to form a circle 3 ±⅛ inch in diameter. The time of application of the paste shall not exceed 15 seconds. This specimen shall be immediately supported, coated side down, on the three balls of the tester and the required weight dropped freely from the specified height, 4.3.4.3.1.1, 4.3.4.3.1.2 and 4.3.4.3.1.3. The weight shall be guided during its fall by a slotted tube, which shall not impede the free fall of the weight, so as to strike the specimen squarely in the center. When the steel cylinder is used, the hemispherical end shall strike the specimen.

4.3.4.3.1.1 In testing ⅛-inch tile, the 0.143-pound weight shall be dropped from the height of 4¼ inches.

4.3.4.3.1.2 In testing 3/16-inch tile, the 0.350-pound weight shall be dropped from a height of 4½ inches.

4.3.4.3.1.3 In testing ¼-inch tile, the 0.350-pound weight shall be dropped from a height of 6 inches.

4.3.4.3.2 The second specimen shall be removed from the bath and tested as described in 4.3.4.3.1, except that it shall be placed on the three balls of the tester so that the grain of the specimen is at right angles to the direction of the grain of the first specimen as placed on the balls.

4.3.4.3.3 The two specimens shall be examined for cracks and breaks. The white coated surface shall be examined closely in good illumination, but not flexed.

4.3.4.4 *Result.*—Two specimens shall be tested. If either specimen breaks or if any cracks are found in either specimen which are of sufficient length to extend beyond the limit of the 3-inch circle of zinc oxide, the tile represented by the sample shall be considered to have failed to meet the requirements of the specification. Such failure shall be cause for rejection.

4.3.5 *Deflection.*

4.3.5.1 *Specimen.*—The specimen shall consist of a piece of tile 2 inches in width and 9 inches long.

4.3.5.2 *Apparatus.*—The apparatus shall be as follows:

4.3.5.2.1 A linoleum knife and straightedge or other suitable means for preparing the specimen.

4.3.5.2.2 *Constant-temperature bath.*—A water bath as described in 4.3.3.1.2.2.

4.3.5.2.3 Two parallel steel rods ¼-inch in diameter and spaced 8 inches from center to center shall be provided for supporting the specimen. A third parallel steel rod ¼ inch

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4.3.6.2.2 Flat slabs of limestone or similar porous material having an absorption of not less than 4 percent or more than 8 percent by weight when immersed in water for 120 hours. The slabs shall be not less than 6¾ x 6¾ x 1¼ inches.

4.3.6.2.3 A water bath of such size that the slabs can be partially immersed to within ¾ to 1 inch of the upper surface.

4.3.6.3 *Procedure.*—The specimen shall be placed on the flat slab of limestone or other porous material. The slab shall be partially immersed in water so that its upper surface is ¾ to 1 inch above the surface of the water. The water shall be maintained at a temperature of 73.5° ±2° F. during the test. The specimen shall be exposed for 120 hours. At the end of the exposure period the amount of curling shall be measured by setting the straightedge on each of the diagonals of the tile and measuring the dip at the center with the dial gage to the nearest 0.001 inch.

4.3.6.4 *Result.* — Two specimens shall be tested. The readings from the two diagonals of each specimen shall be averaged. If the average of the two values from either specimen exceeds the requirements of the specification, the tile represented by the sample shall be considered to have failed to meet the requirements of the specification. Such failure shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Packing.—Unless otherwise specified, the subject commodity shall be prepared for shipment to permit acceptance by common or other carriers for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.2 Marking.—Unless otherwise specified, shipping containers shall be marked with the name of the material as defined by the contract or order under which the shipment is made, the size, thickness, color, the quantity of each thickness contained therein, the name of the manufacturer, the name of the contractor, the number of the contract or order, the name and address of the consignee, and the gross weight.

6. NOTES.

6.1 Ordering data.—Purchasers should exercise any desired options offered herein (see 1.2, 1.3, 3.4, 3.5, 5.1, 5.2, and sec. 7), and should cover the following particulars in the invitation for bids and purchase orders:

6.1.1 Thickness and size (1.2, 3.4, 3.5, and 6.3).

6.1.2 *Color, finish, and surface characteristics.*—Specific color numbers of the manufacturers or purchasing agency should be specified when samples are required with bid for color, finish, and surface characteristics. (See 3.3 and 6.4.)

6.1.2.1 In normal manufacture, tile varies somewhat in hue and shade and are subject to minor changes due to age and exposure. A variation in marbelizing and mottling is characteristic of the material.

6.2 Tile covered by this specification is not oil-resistant and is not necessarily acid-resistant. Where especially severe service conditions are to be encountered, special specifications should be prepared giving the particular requirements to be met.

6.3 Sizes.—Asphalt tile is commercially available in the following sizes in inches:

Square: 9 by 9, and 12 by 12.
Rectangular: 18 by 24.

6.4 Colors.—Asphalt tile is commercially available in a wide variety of colors, although few manufacturers produce all colors in all sizes. Colors are classified by manufacturers into 4 groups: A, B, C and D. Group A are dark plain colors; group B marbelized colors with dark background; groups C and D plain and marbelized colors in intermediate and lighter shades. Normally, cost varies upward from group A to group D.

6.5 Accessory materials.

6.5.1 *Cove base.*—Asphalt cove bases are available in 4- and 6-inch heights. The requirements of this specification are not considered applicable to asphalt cove base.

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Military Specifications:

JAN–P–105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.

JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.

MIL–B–107—Boxes, Wood, Wirebound (Overseas Type).

JAN–P–108—Packaging and Packing for Overseas Shipment—Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.

Navy Department Specification:

General Specifications for Inspection of Material.

Military Standard:

MIL–STD–129—Marking of Shipments.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.3.2 *Inspection procedures.*—For Naval purchases, for general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

7.3.3 *Packaging.*—Unless otherwise specified, commercial packaging will be acceptable.

7.3.4 *Packing.*

7.3.4.1 *For domestic shipment.*—The tiles, packaged as described in 7.3.3, shall be packed in wood-cleated fiberboard, wood-cleated plywood, nailed wood, wirebound wood, corrugated fiber, or solid fiber boxes, conforming to Federal Specifications NN–B–591, NN–B–601, NN–B–621, NN–B–631, LLL–B–631, and LLL–B–636, respectively. The gross weight of wood boxes shall not exceed approximately 200 pounds. The gross weight of fiberboard boxes shall be subject to the limitation of the box specification.

7.3.4.2 *For overseas shipment.*—The tiles, packaged as described in 7.3.3, shall be packed in cleated plywood, nailed wood, wirebound wood or fiberboard boxes, conforming to Military Specifications JAN–P–105, JAN–P–106, MIL–B–107, and JAN–P–108, respectively. The gross weight of wood boxes shall not exceed approximately 200 pounds, and of fiberboard boxes approximately 70 pounds.

7.3.5 *Marking.*—In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

7.3.6 *Ordering data.*—Procurement documents should specify whether domestic or overseas packing and marking are required. (See 7.3.4 and 7.3.5.)

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7.4 Air Force.

7.4.1 *Applicable standard.*

Military Standard:

MIL–STD–129—Marking of Shipments.

7.4.2 *Marking.*—Interior packages and exterior shipping containers shall be marked in accordance with Military Standard MIL–STD–129. The nomenclature shall be as follows: Tile; Floor, Asphalt, Material_____ (if solid), Thickness_____, Width_____ Length_____, Color_____, Background Color_____ (if applicable), Federal Specification SS–T–306b.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

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